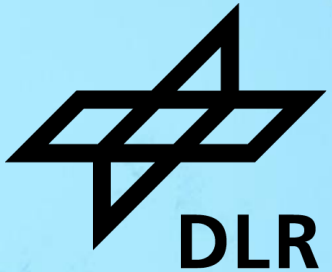
RPCHI: Practical Part Introduction

Organization & Kuka Sunrise Introduction

Arne Sachtler, Maximilian Mühlbauer
8th January 2024



Organisational

☐	*Group A* *Hinweis: Auf Datum klicken, um Einzeltermin zu verschieben. In der Spalte Serie auf S klicken, um Terminserie zu verschieben.*									
☐	Di	05.12.2023	09:00	12:00	03.09.034, Praktikumsraum (5609.03.034)	Abhaltung	fix		Simulation	**Sachtler**, Arne, M.Sc.
☐	Di	12.12.2023	09:00	12:00	03.09.034, Praktikumsraum (5609.03.034)	Abhaltung	fix		Simulation	**Sachtler**, Arne, M.Sc.
☐	Di	19.12.2023	09:00	12:00	03.09.034, Praktikumsraum (5609.03.034)	Abhaltung	fix		Simulation	**Sachtler**, Arne, M.Sc.
☐	Di	09.01.2024	09:00	12:00	03.07.011, Praktikum/Labor/Mob. (5607.03.011)	Abhaltung	geplant			**Mühlbauer**, Maximilian, M.Sc. **Sachtler**, Arne, M.Sc.
☐	Di	16.01.2024	09:00	12:00	03.07.011, Praktikum/Labor/Mob. (5607.03.011)	Abhaltung	geplant			**Mühlbauer**, Maximilian, M.Sc. **Sachtler**, Arne, M.Sc.
☐	Di	23.01.2024	09:00	12:00	03.07.011, Praktikum/Labor/Mob. (5607.03.011)	Abhaltung	geplant			**Mühlbauer**, Maximilian, M.Sc. **Sachtler**, Arne, M.Sc.
☐	Di	30.01.2024	09:00	12:00	03.07.011, Praktikum/Labor/Mob. (5607.03.011)	Abhaltung	geplant			**Mühlbauer**, Maximilian, M.Sc. **Sachtler**, Arne, M.Sc.
☐	Di	06.02.2024	09:00	12:00	03.07.011, Praktikum/Labor/Mob. (5607.03.011)	Abhaltung	geplant			**Mühlbauer**, Maximilian, M.Sc. **Sachtler**, Arne, M.Sc.

- 5 weeks.
- Groups B & D: Now starting at 13:15 instead of 13:00

Groups within Groups

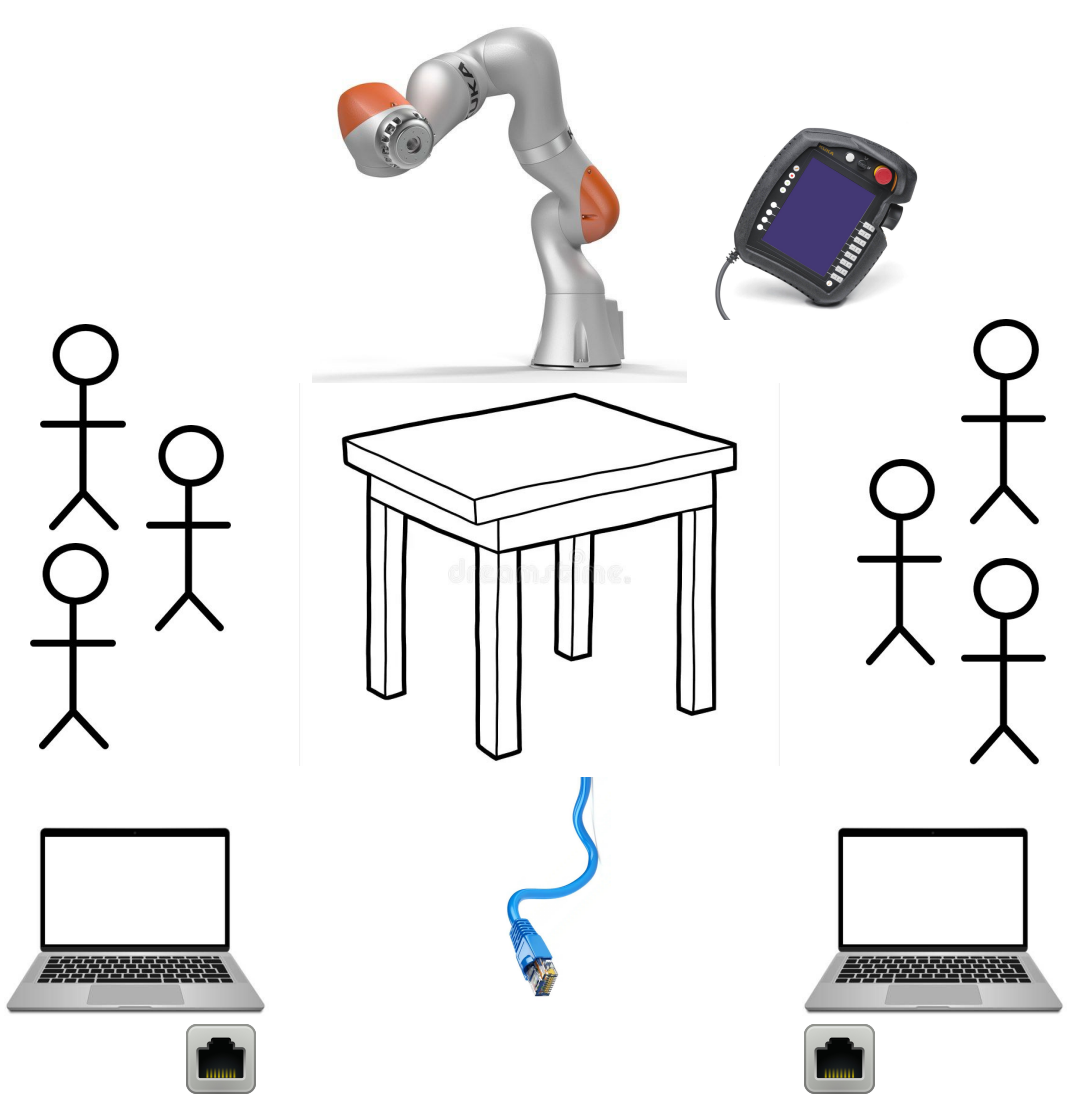
- Within each group (A, B, C, D) please form four groups of three people.
- This group of three will
  - work on the robot exercises together,
  - do the oral examination presentation together,
  - stay together as one group for the remainder of this course.
- Tell us your group today or right at the beginning of the first practical part
- Suggestions for good performing groups:
  - be interdisciplinary: mix between computer science / robotics / mechanical eng.
  - people should have worked on different problems (IK / NO / FCI)

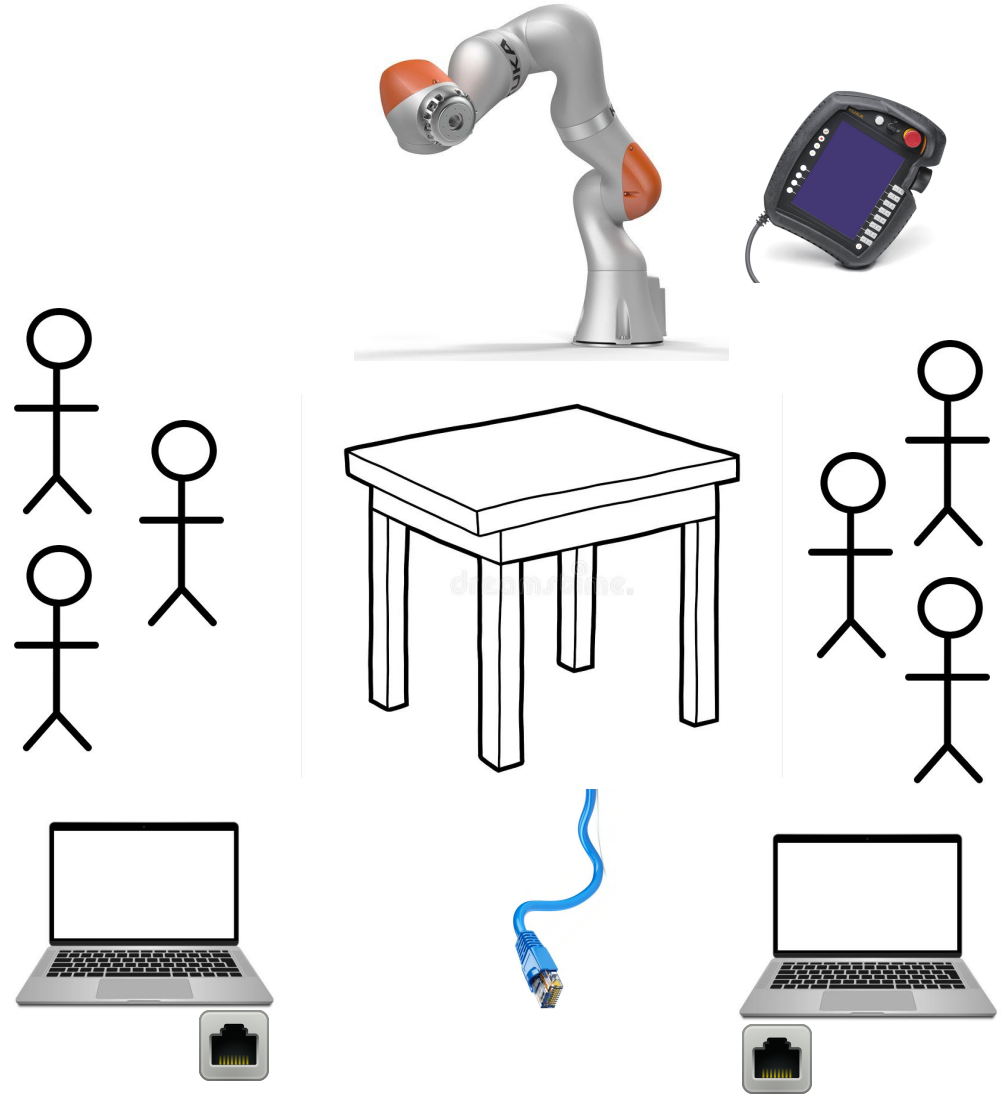
Examination

- In your group of three
- Showing results of the practical part
  - You decide what to show!
  - If you want you may also show simulation results (no requirement)
- 5 mins presentation + 5 mins questions per team member (probably)
  - you make sure you distribute time properly
  - questions are regarding…
    - what you have shown in the presentation
    - might be about the simulation part
    - might be about lecture content
  - we want to see that you understand what's going on
- Remember to register in TUMonline!
- Preliminary dates:
  - Groups A & B: Wednesday, 21.02.2023
  - Groups C & D: Wednesday, 28.02.2023

Lab Setup

Be fair and distribute the time on the robot equally among the two groups.

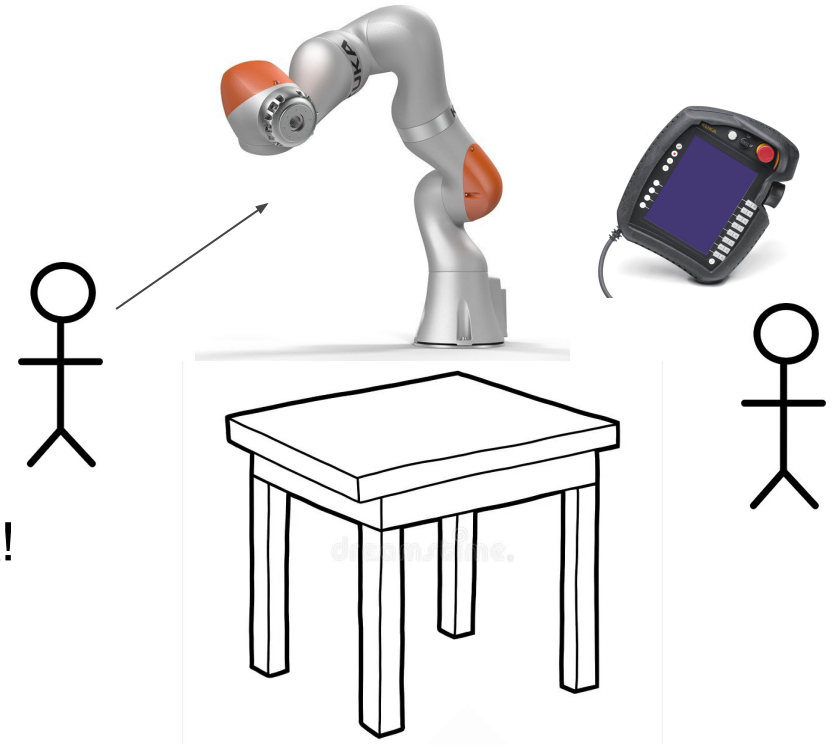




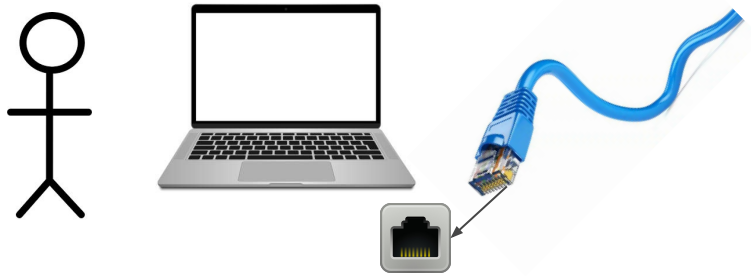
How to work on the Robot

Groups of three, because

- 1 Person interacts with the robot
  - some tasks need physical interaction
- 1 Person controls the software
- 1 Person operates the SmartPad
  - presses the “allow” button
  - must observe operator and robot
  - never press “allow” when not looking at the robot!



- You have one robot for two groups.



Software Setup

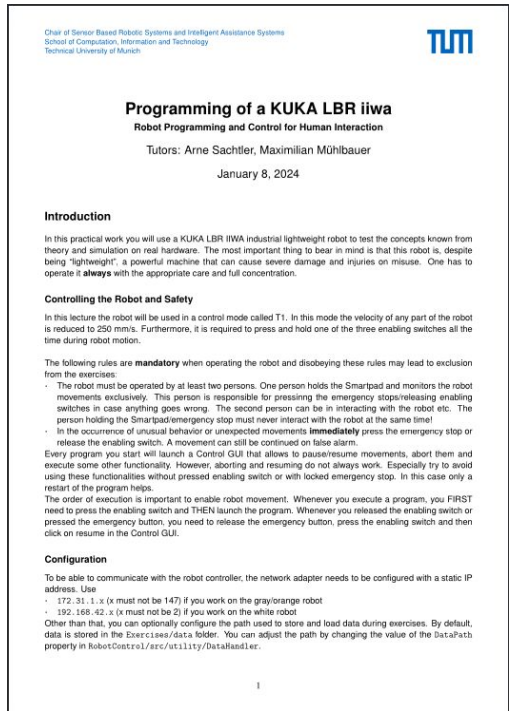
- Per group you need at least one computer with physical ethernet port.
- Set up a static IPv4 network
  - Subnetmask: `255.255.255.0`
  - Robot IP: `172.31.1.147`
  - Your IP: `172.31.1.x`, where `x` is **not** `147`
- Install a Java IDE. We recommend: IntelliJ
  - Community Edition is more than enough
- Fork the repository for this course once for your group.
  - `https://gitlab.lrz.de/arne/iiwa-exercises`
- Clone it directly to IntelliJ

Material you will find in Moodle

GitLab

Link to the template repo

Tutorial



Chair of Sensor Based Robotic Systems and Intelligent Assistance Systems
School of Computation, Information and Technology
Technical University of Munich

Programming of a KUKA LBR iiwa

Robot Programming and Control for Human Interaction

Tutors: Arne Sachtler, Maximilian Mühlbauer

January 8, 2024

Introduction

In this practical work you will use a KUKA LBR IIWA industrial lightweight robot to test the concepts known from theory and simulation on real hardware. The most important thing to bear in mind is that this robot is, despite being "lightweight", a powerful machine that can cause severe damage and injuries on misuse. One has to operate it **always** with the appropriate care and full concentration.

Controlling the Robot and Safety

In this lecture the robot will be used in a control mode called T1. In this mode the velocity of any part of the robot is reduced to 250 mm/s. Furthermore, it is required to press and hold one of the three enabling switches all the time during robot motion.

The following rules are **mandatory** when operating the robot and disobeying these rules may lead to exclusion from the exercises:

- The robot must be operated by at least two persons. One person holds the Smartpad and monitors the robot movements exclusively. This person is responsible for pressing the emergency stops/releasing enabling switches in case anything goes wrong. The second person can be in interacting with the robot etc. The person holding the Smartpad/emergency stop must never interact with the robot at the same time!
- In the occurrence of unusual behavior or unexpected movements **immediately** press the emergency stop or release the enabling switch. A movement can still be continued on false alarm.

Every program you start will launch a Control GUI that allows to pause/resume movements, abort them and execute some other functionality. However, aborting and resuming do not always work. Especially try to avoid using these functionalities without pressed enabling switch or with locked emergency stop. In this case only a restart of the program helps.
The order of execution is important to enable robot movement. Whenever you execute a program, you FIRST need to press the enabling switch and THEN launch the program. Whenever you released the enabling switch or pressed the emergency button, you need to release the emergency button, press the enabling switch and then click on resume in the Control GUI.

Configuration

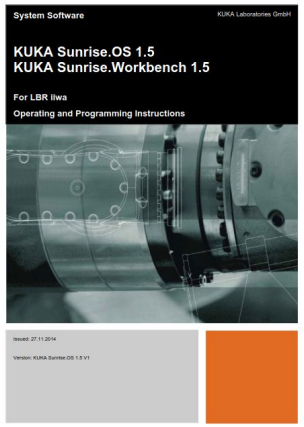
To be able to communicate with the robot controller, the network adapter needs to be configured with a static IP address. Use

- `172.31.1.x` (x must not be 147) if you work on the gray/orange robot
- `192.168.42.x` (x must not be 2) if you work on the white robot

Other than that, you can optionally configure the path used to store and load data during exercises. By default, data is stored in the `Exercises/data` folder. You can adjust the path by changing the value of the `DataPath` property in `RobotControl/src/utility/DataHandler`.

1

KUKA



System Software

KUKA Laboratories GmbH

KUKA Sunrise.OS 1.5
KUKA Sunrise.Workbench 1.5

For LBR iiwa

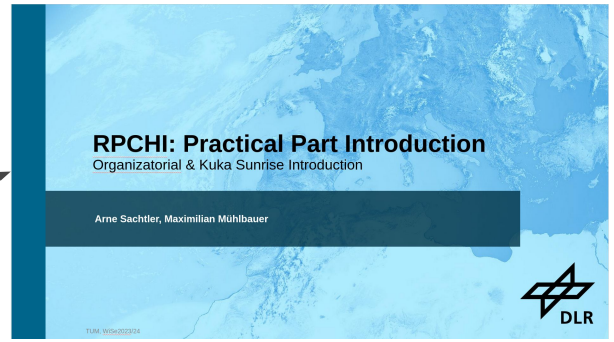
Operating and Programming Instructions

Issued: 27.11.2014

Version: KUKA Sunrise.OS 1.5 V1

Robot documentation

These slides



RPCHI: Practical Part Introduction

Organizatorial & Kuka Sunrise Introduction

Arne Sachtler, Maximilian Mühlbauer

DLR

TUM: WiSe2023/24

Plan for the first session

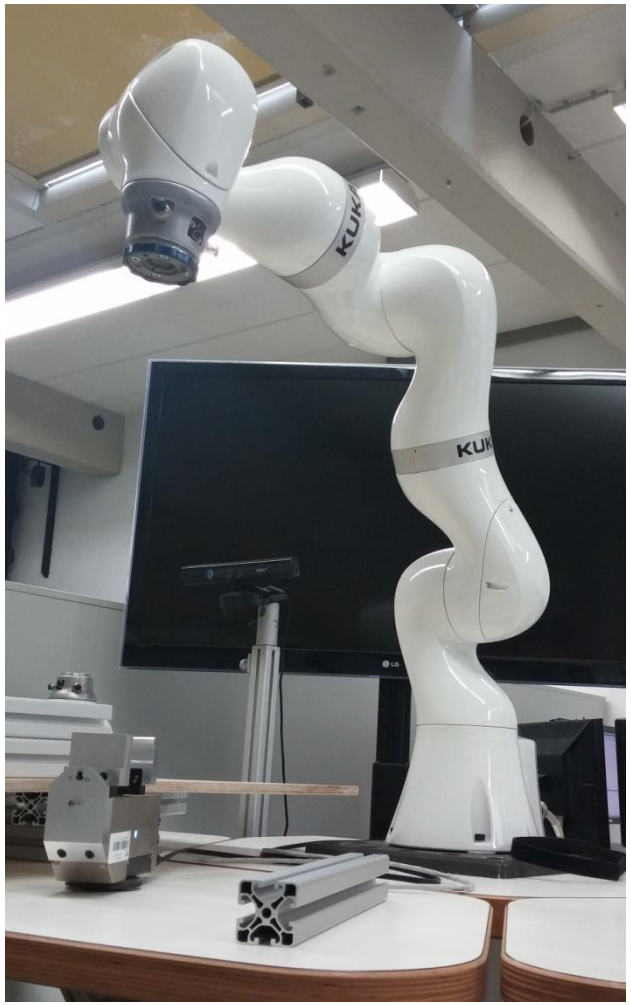
- Tomorrow (A & B) or Wednesday (C & D)
- We start with a security briefing:
  - Please don’t be late.
  - You must attend the security briefing and sign that you heard it.
  - Do not work on the robot before you have signed that you got the security briefing!
  - After you have signed, you may carefully start working with the robot.
  - This is very important! (because of insurance, legal stuff)
- We advise you to
  - Form groups already
  - Install the software, prepare at least one computer per group
  - Otherwise you might waste time with downloading / group forming



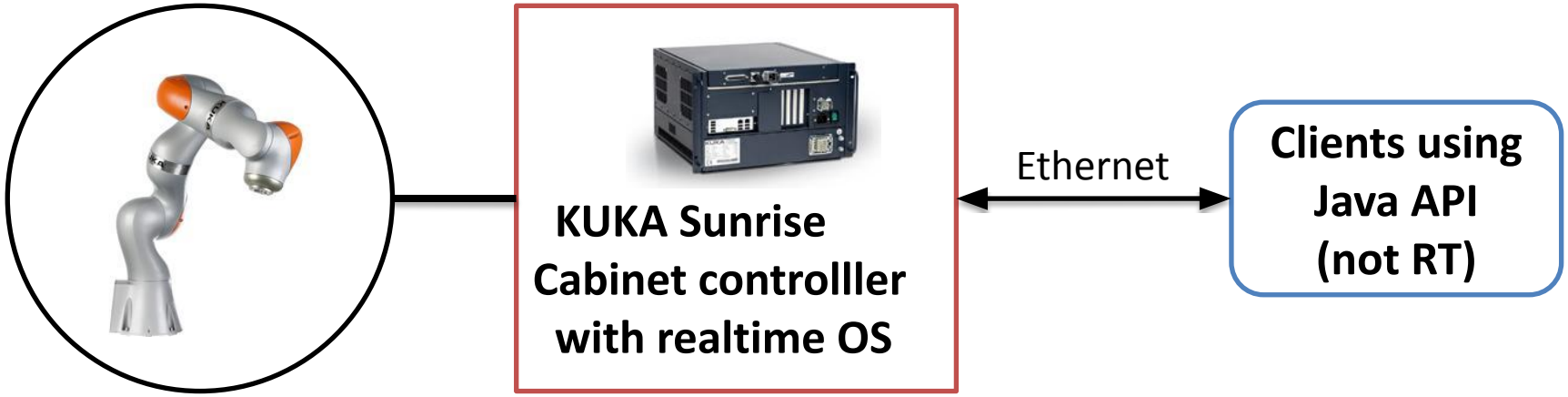
Programming with KUKA Sunrise.OS

KUKA LBR iiwa 7 R800

- 7 DOF lightweight robot
- Weight: 22.5 kg
- Load: 7 kg
- Height: 1.25 m
- Range: 0.8 m



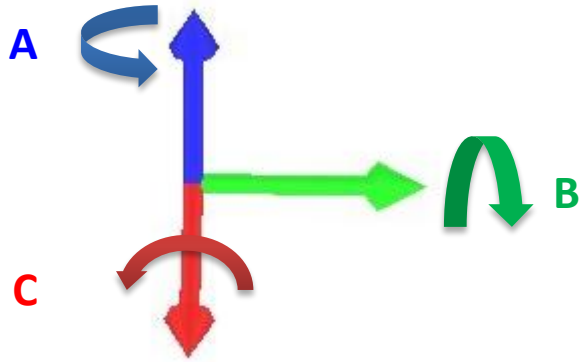
System Setup

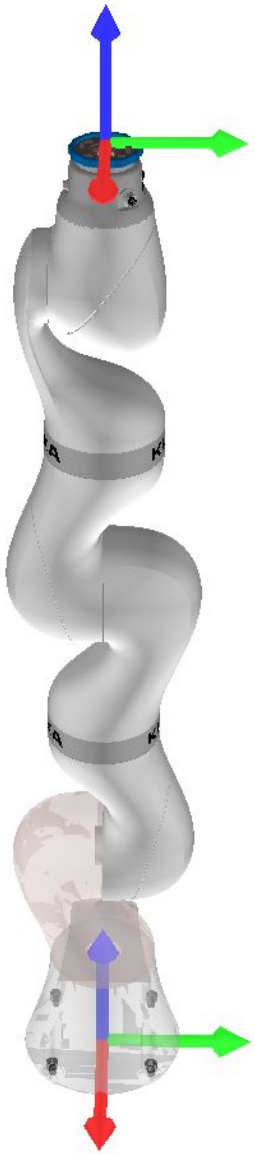


- Control Loop run on the Cabinet, not on your Computer.
- You basically only parameterize and select controller using the Java API.

Coordinate Systems

- World CS located in base of robot
- KUKA convention:
  - Distances in mm
  - Rotations in degree (frames)
  - **or** radians (joint positions)
- KUKA Euler angle convention:





Basic Usage

- Important classes:
  - **LBR**: communication with robot, created at startup
  - **JointPosition**: 7D vector of motor positions [rad]

```
j = new JointPosition(0,Math.Pi,0,0,0,0,0)
```

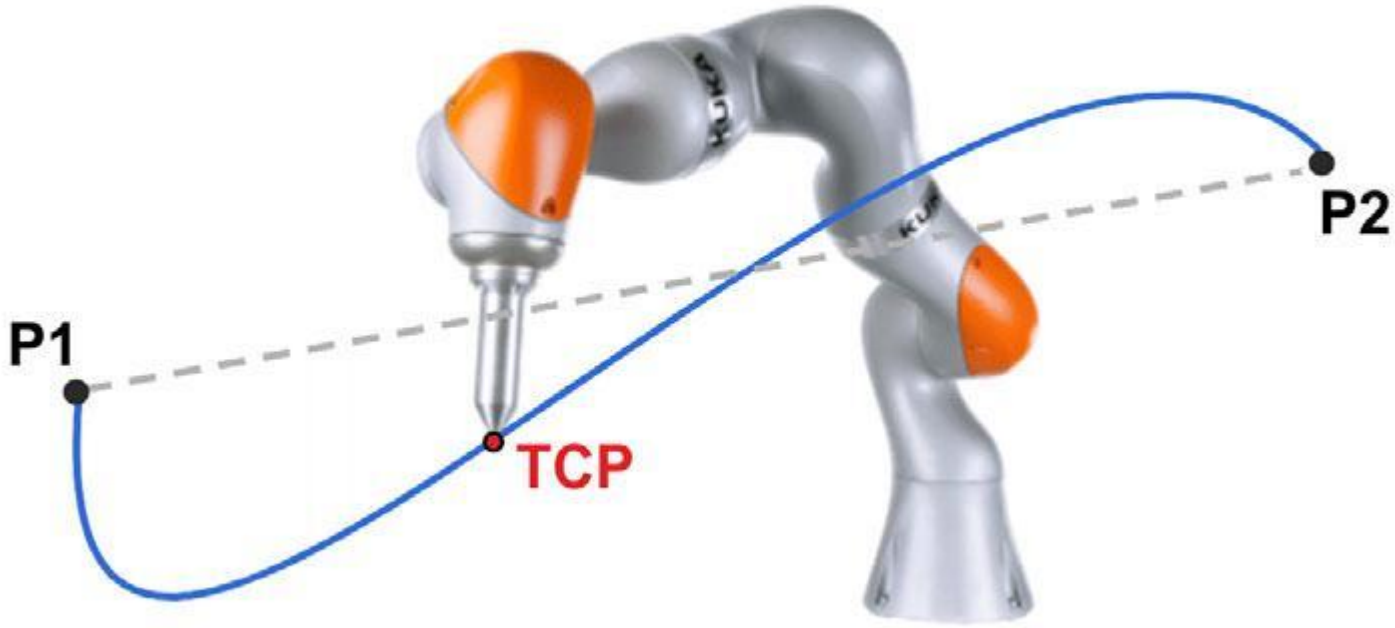
  - **Frame**: describes spatial displacements in space, contains redundancy parameters

```
f = new Frame (x,y,z,a,b,c)
```

  - **IMotion**: Interface for motion commands

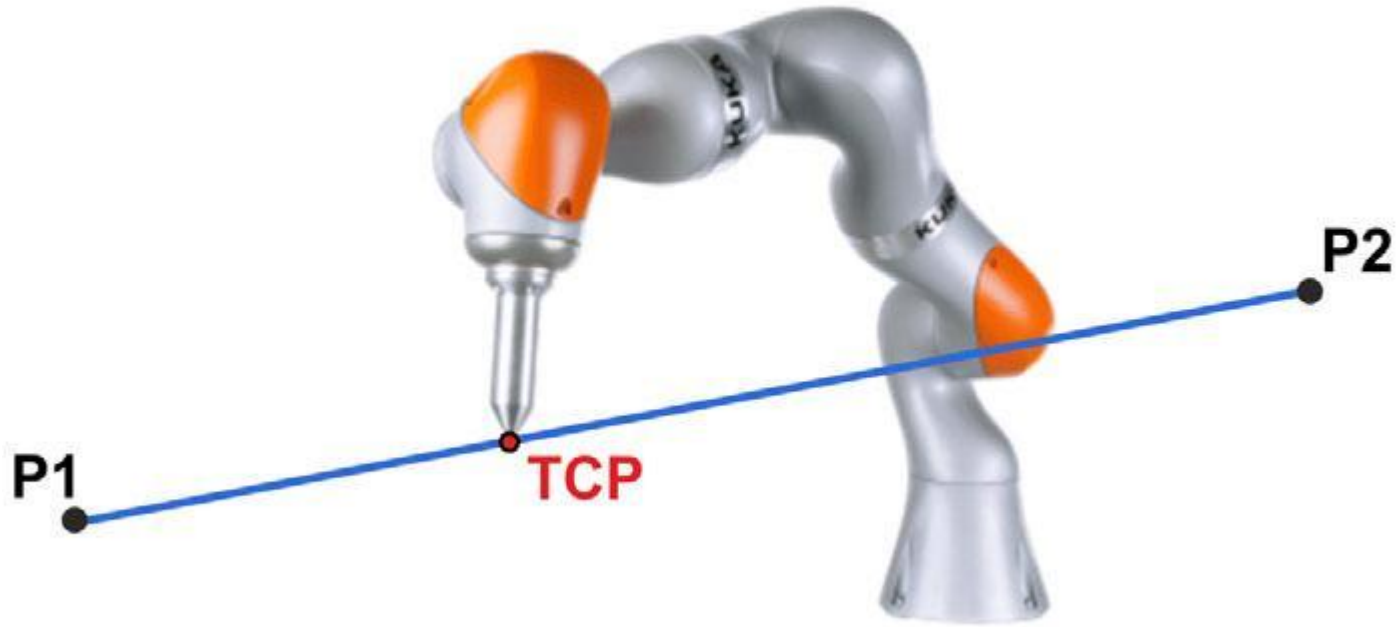
Basic Motions: PTP

```
BasicMotions.ptp(JointPosition);
BasicMotions.ptp(Frame);
```



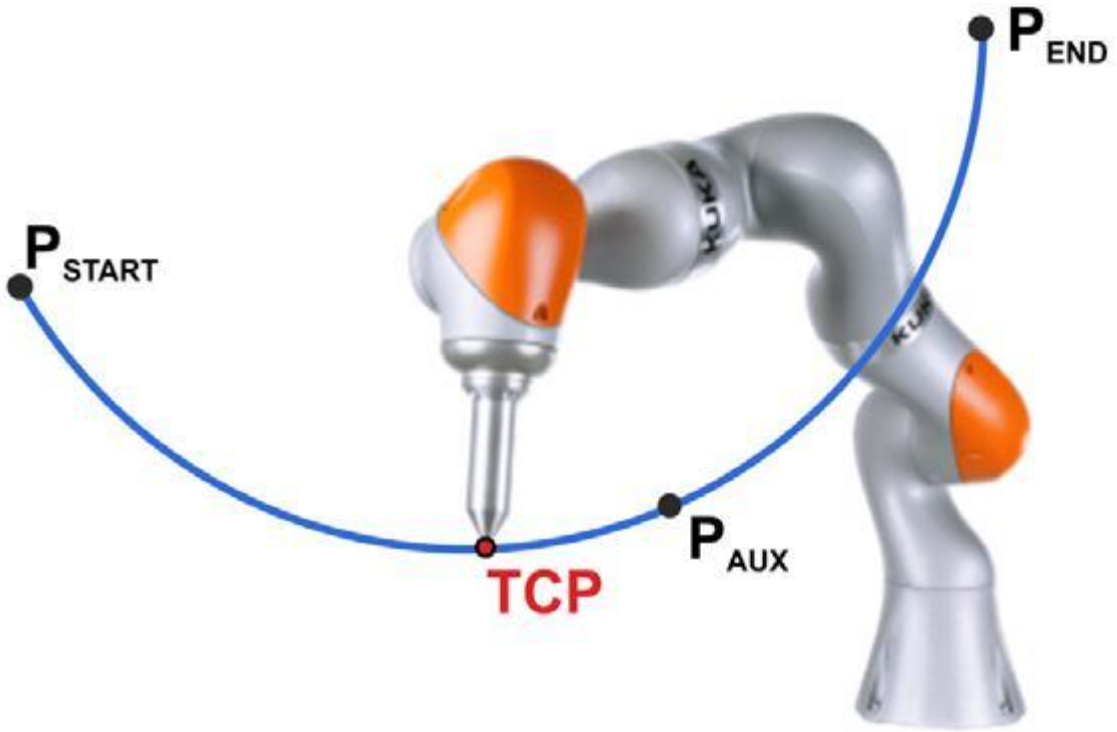
Basic Motions: LIN

```
BasicMotions.lin(Frame);
BasicMotions.linRel(x, y, z);
```



Basic Motions: CIRC

```
BasicMotions.circ(Frame aux, Frame end);
```



Basic Motions

- Execute motions synchronously/async.:

```
LBR.move(IMotion); // blocking call
LBR.moveAsync(IMotion); // non-blocking call
```

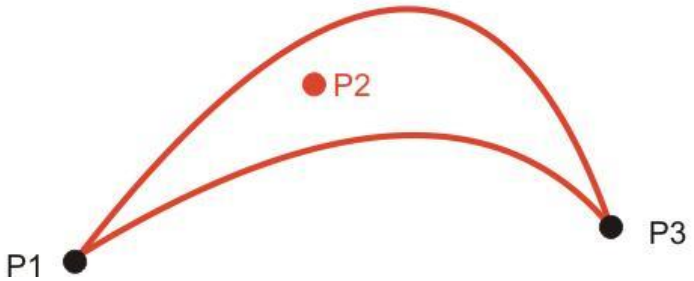
- Important parameters:
  - Velocity
  - Acceleration

```
IMotion.setJointAccelerationRel(0.5).setJointVelocityRel(0.3);
```

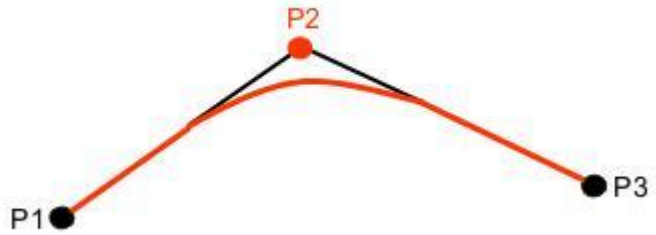
Blending

```
IMotion.setBlendingRel(0.2); // relative value
IMotion.setBlendingCart(20); // in millimeters
```

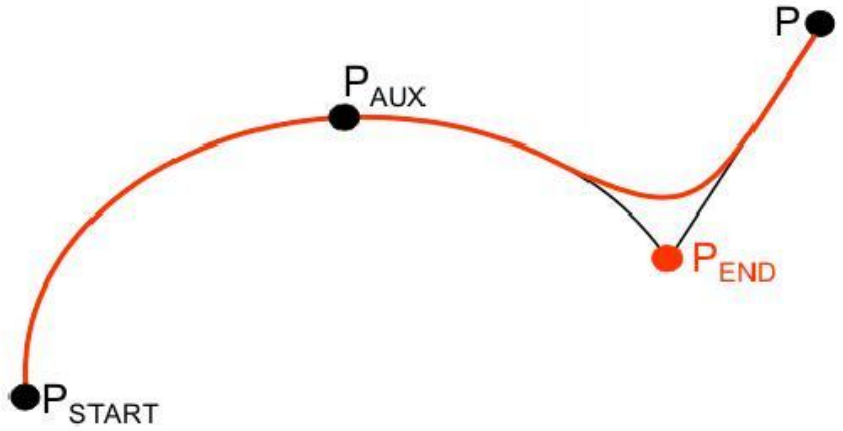
PTP:



LIN :



CIRC:



Combine Motions

- Create motion batches to combine motions

```
Vector<RobotMotion<?>> l = [...];
l.add(IMotion m1);
l.add(IMotion m2);
...
MotionBatch mb = new MotionBatch(l);
```

- Set parameters per motion or for complete motion batch
- Blending of motion commands (for motion batches or async movements)

Read Sensor Values

- Joint positions and TCP frame

```
LBR.getCurrentJointPosition();
LBR.getCurrentCartesianPosition(LBR.getFlange());
```

- Torque sensors

```
LBR.getSensorForMeasuredTorque().getSensorData();
LBR.getSensorForExternalTorque().getSensorData();
LBR.getSensorForExternalForce().getSensorData();
```

Conditions

- Place break conditions on motions:
  - JointTorque Condition
  - Force Condition

```
IMotion.breakWhen(new JointTorqueCondition(  JointEnum, minTorque, maxTorque))
```

- Combine conditions with and, or, xor..

```
ICondition cond = ICondition.and(ICondition);
```

Compliant Control

- Joint impedance and Cartesian impedance controller available

```
IMotion.setMode(ImpedanceControlMode);
```

- Joint impedance controller:

```
JointImpedanceControlMode JICM = new Joint…();
JICM.setStiffness(<7 values>);
JICM.setDamping(<7 values>);
...
```

Compliant Control II

- Cartesian impedance controller:
  - DOF-dependent (stiffness, damping , maximum TCP force, …)

```
CartesianImpedanceControlMode CICM = new Cart…();
CICM.parametrize(CartDOF.X).setStiffness(3000);
CICM.parametrize(CartDOF.ALL).setDamping(0.7);
CICM.setMaxControlForce(maxForceX, maxForceY, ...);
```

  - DOF-independent (nullspace stiffness, -damping)

```
CICM.setNullSpaceStiffness(10.0);
CICM.setNullSpaceDamping(0.7);
```